

BARCELONA, Venezuela

WMO: 804190

Lat: 10.107N

Lon: 64.689W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (W04)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.3	22.0	14.7	10.4	29.5	15.9	11.3	28.7	10.2	32.0	9.6	32.0	0.4	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.8	35.8	26.1	35.1	25.9	34.7	25.8	29.0	33.0	28.2	32.4	27.7	31.9	5.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.2	31.5	27.2	22.9	30.3	26.8	22.5	29.9	94.7	32.6	90.7	32.3	88.1	32.6	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.3	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	27.6	27.8	28.5	28.9	29.3	28.7	28.1	28.4	28.8	28.9	28.8	28.1
	DBStd	1.01	0.86	0.85	0.94	1.00	0.87	0.98	0.94	0.80	0.93	0.86	0.72	0.86
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6752	547	498	572	567	597	562	563	569	564	586	565	562
	CDD18.3	3710	288	264	314	317	339	312	304	311	314	328	315	304
	CDH23.3	40939	3071	2876	3644	3764	4084	3454	3086	3178	3372	3618	3519	3272
	CDH26.7	18762	1358	1313	1748	1798	1983	1561	1307	1353	1559	1691	1628	1461
Wind	WSAvg	2.1	2.1	2.4	2.5	2.7	2.3	1.9	1.7	1.8	2.0	2.1	1.7	1.9
Precipitation	PrecAvg	610	6	3	4	8	40	91	121	117	76	58	53	24
	PrecMax	938	41	44	30	84	226	219	295	296	208	154	201	70
	PrecMin	400	0	0	0	0	0	9	52	16	13	8	4	1
	PrecStd	133	10	7	7	13	43	44	56	57	39	36	47	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.1	36.0	36.0	36.0	35.2	35.0	34.9	36.1	36.0	35.2	35.1
		MCWB	25.1	26.0	25.6	26.7	25.8	25.7	26.0	25.9	26.6	26.2	26.4	25.6
	2%	DB	33.9	34.0	34.9	35.0	35.1	34.3	33.9	34.0	35.0	35.0	34.8	34.1
		MCWB	25.1	25.4	25.4	25.7	26.0	25.7	25.7	25.9	26.0	26.1	26.4	25.5
	5%	DB	33.0	33.1	34.0	34.1	34.2	33.9	33.0	33.1	34.1	34.1	34.0	33.2
		MCWB	25.0	24.8	25.3	25.1	25.7	25.7	25.4	25.8	26.0	26.3	26.2	25.1
	10%	DB	32.1	32.2	33.1	33.2	33.8	32.9	32.1	32.2	33.2	33.3	33.1	32.7
		MCWB	24.8	24.4	24.9	25.0	25.6	25.5	25.3	25.6	25.8	26.0	26.0	25.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.6	28.9	29.6	29.6	29.0	28.3	28.1	28.7	28.8	30.6	28.0
		MCDB	31.9	33.3	32.9	34.2	34.0	32.5	32.0	31.8	32.3	32.7	32.6	31.7
	2%	WB	27.1	27.1	27.8	28.1	28.4	27.8	27.3	27.2	27.5	27.9	28.2	27.1
		MCDB	31.4	31.5	32.2	32.7	33.0	31.3	30.9	31.1	32.2	32.2	32.1	31.1
	5%	WB	26.4	26.3	26.8	26.8	27.4	27.0	26.7	26.6	26.9	27.2	27.3	26.5
		MCDB	30.6	30.6	31.3	31.5	31.9	30.8	30.4	30.6	31.6	32.1	31.6	30.8
	10%	WB	25.8	25.6	26.2	26.2	26.7	26.4	26.2	26.2	26.5	26.7	26.7	26.1
		MCDB	29.9	30.1	30.6	30.7	31.0	30.2	29.8	30.2	31.1	31.5	31.0	30.4
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.6	10.5	9.9	9.8	9.5	9.3	9.4	10.1	10.0	9.9	10.0
		MCDBR	11.1	11.2	11.2	10.6	10.4	10.4	10.0	10.0	10.9	10.8	10.6	10.6
		MCWBR	4.5	4.5	4.5	4.0	4.0	3.6	3.8	3.8	4.0	4.1	4.2	4.0
	5% WB	MCDBR	10.6	10.8	10.6	10.1	10.1	9.6	9.5	9.6	10.4	10.1	10.1	10.2
		MCWBR	5.3	5.5	5.4	4.8	5.1	4.4	4.3	4.2	4.5	4.6	5.1	4.4
Clear-Sky Solar Irradiance	taub		0.379	0.398	0.407	0.459	0.449	0.479	0.451	0.424	0.413	0.393	0.389	0.394
	taud		2.561	2.495	2.469	2.321	2.376	2.268	2.374	2.468	2.479	2.552	2.569	2.544
	Ebn at Noon		930	921	913	855	848	812	839	874	894	914	914	907
	Edh at Noon		101	111	116	133	123	135	122	113	113	104	100	101
All-Sky Solar Radiation	RadAvg		5.37	5.90	6.15	6.01	5.77	5.59	5.72	5.97	6.08	5.71	5.23	5.05
	RadStd		0.20	0.22	0.39	0.32	0.33	0.24	0.24	0.23	0.19	0.17	0.27	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page